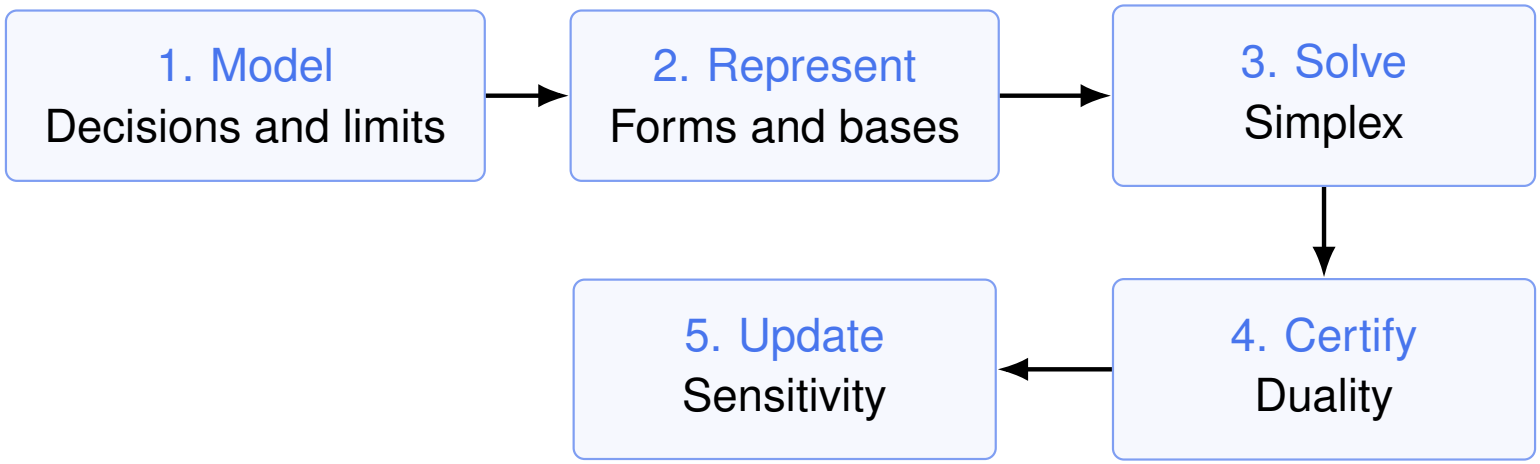
Operations Research

Midterm Review: Lectures 1–5

The route through this review



One production example will connect the methods. Small additional examples will clarify the special cases.

The aim is to know **what each concept means**, **when to use it**, and **what justifies the conclusion**.

Modeling: translate a decision into mathematics

Variables	Objective	Constraints
What can we choose?	What measures quality?	What is allowed?

$$\max\ f(x) \qquad \text{subject to} \quad g_i(x) \le 0, \quad x \in \mathcal{D}.$$

Define every variable **in words**, with its unit and domain.

An **LP** has a linear objective, linear constraints and continuous variables. Integer restrictions give an **IP**; a nonlinear expression may give an **NLP**.

Example: $x_1 x_2$ is nonlinear; $3x_1 + 2x_2$ is linear.

A production example we will reuse

Let $x_1, x_2 \geq 0$ be the daily amounts of divisible products P and Q. All output sells.

Per production unit	P	Q	Available
Resource 1 (hours)	2	1	10 h
Resource 2 (hours)	1	2	8 h
Net profit (MAD)	500	400	–

Measure profit in **hundreds of MAD**:

$$z = \frac{500x_1 + 400x_2}{100}.$$

Only the unit of the objective changes. The resource constraints keep their units.

$$\begin{aligned} \max \quad & z = 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10, \\ & x_1 + 2x_2 \leq 8, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Canonical form and equational form

For our maximization convention, the canonical form is

$$\max\{c^T x : Ax \leq b,\ x \geq 0\}.$$

An equational form uses **equalities and nonnegative variables**:

$$\max\{c^T x : Ax = b,\ x \geq 0\}.$$

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Original constraint	Equality	Meaning
$a^T x \leq \beta$	$a^T x + s = \beta$	$s \geq 0$: unused amount
$a^T x = \beta$	unchanged	no slack needed

The original decisions are unchanged; the equational vector includes the new slacks.

Make the variable domains fit the form

A variable with a lower bound

For $-2 \leq t \leq 3$, set $u = t + 2$. Then

$$\boxed{t = u - 2, \qquad 0 \leq u \leq 5.}$$

Substitute $u - 2$ everywhere t occurs.

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Original domain	Nonnegative variables
$t \le 0$	$t = -u$, $u \ge 0$
$t \in \mathbb{R}$ unrestricted	$t = u - v$, $u, v \ge 0$

For a minimization objective, maximizing its negative reverses the objective sense. Track any constant introduced by a substitution when reporting the value.

Matrix notation: order the variables first

For the production model, add $s_1, s_2 \geq 0$ and define

$$x = (x_1, x_2, s_1, s_2)^{\mathsf{T}}.$$

In **this order**,

$$A = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 10 \\ 8 \end{pmatrix}, \qquad c = \begin{pmatrix} 5 \\ 4 \\ 0 \\ 0 \end{pmatrix}.$$

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In general, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $x, c \in \mathbb{R}^n$:

$$(Ax)_i = \sum_{j=1}^{n} a_{ij} x_j, \qquad c^{\mathsf{T}} x = \sum_{j=1}^{n} c_j x_j.$$

The **rows** represent equations; the **columns** correspond to variables.

A redundant inequality does not restrict the decisions

An inequality is **redundant** if the other constraints already imply it. For example,

$$\begin{aligned} & 2x_1 + x_2 \leq 10, \\ + \quad & x_1 + 2x_2 \leq 8 \\ \hline & 3x_1 + 3x_2 \leq 18. \end{aligned}$$

Thus adding $3x_1 + 3x_2 \leq 18$ does not change the feasible set.

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Thus adding $3x_1 + 3x_2 \le 18$ does not change the feasible set.

A useful sufficient test is to combine other inequalities with **nonnegative weights** and derive the candidate inequality.

Removing it before adding slacks reduces the number of variables. Keeping it is valid and gives the **same optimal value**.

Equality systems: consistency and rank

For a general equality system, apply elimination to $[A \mid b]$.

What elimination gives	What it means
$0 = 0$	This equation can be removed.
$0 = \beta$, $\beta \neq 0$	The equality system is inconsistent.
r pivots in the coefficient block A	$\operatorname{rank}(A) = r$.

If the system is consistent, keep r independent rows and then select r **independent columns** to form a basis matrix.

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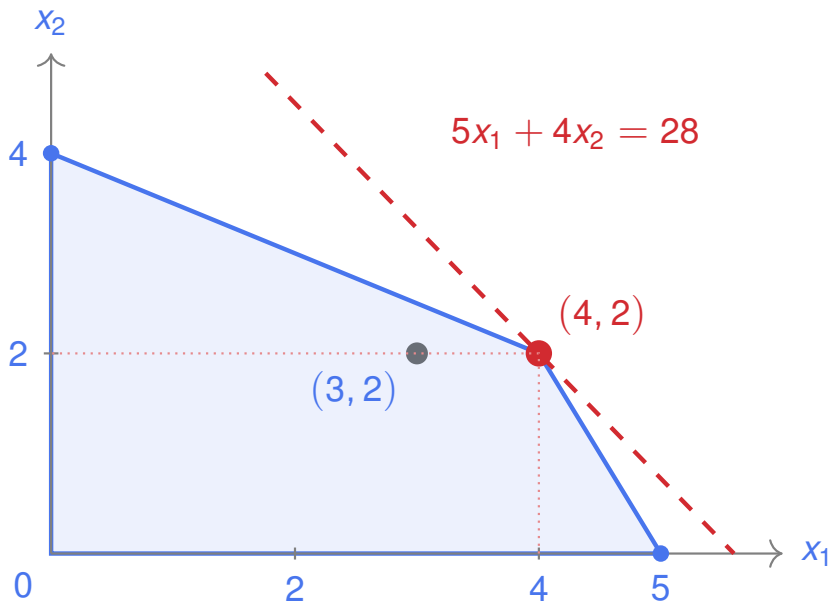
If the system is consistent, keep r independent rows and then select r **independent columns** to form a basis matrix.

In our running example, A has two rows and contains I_2 in its slack columns. Hence

$$\operatorname{rank}(A) \leq 2, \qquad \operatorname{rank}(A) \geq 2 \quad \Longrightarrow \quad \boxed{\operatorname{rank}(A) = 2.}$$

From feasible plans to finite candidates

Feasible and optimal are different claims



A **feasible** plan respects every constraint.

An **optimal** plan has the best objective among all feasible plans.

Here $(3, 2)$ is feasible and earns 23. The plan $(4, 2)$ earns 28.

A graphical answer should identify the region, its vertices and the optimal level, then **state the conclusion in words**.

An LP has three possible outcomes

Outcome	Meaning	Example
Finite optimum	A best feasible value exists	Our production LP
Unbounded objective	Feasible values grow without limit	$\max x_1$, $x_1 \geq 0$
Infeasible	No plan satisfies all constraints	$x_1 \geq 2$, $x_1 \leq 1$

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An **unbounded feasible set** can still have a finite optimum:

$$\max\{-x_1 : x_1 \geq 0\} \quad \text{has optimum} \quad 0.$$

The relevant question is whether the **objective values** are bounded above.

A basis and its basic feasible solution

Consider $\max\{c^T x : Ax = b,\ x \geq 0\}$, with $\text{rank}(A) = m$. Choose m **independent columns**, in a specified order:

$$B = [A_{j_1} \ \cdots \ A_{j_m}] \in \mathbb{R}^{m \times m}.$$

The corresponding basic solution is

$$\boxed{x_{\mathcal{N}} = 0, \qquad x_B = B^{-1} b.}$$

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It is a **BFS** exactly when $B^{-1} b \geq 0$. It is **nondegenerate** when all m basic coordinates are strictly positive.

A BFS has at most m nonzero coordinates. Basic coordinates may also be zero. These statements concern the **full equational vector**, including slacks.

Example: compute the point from the chosen columns

For $x = (x_1, x_2, s_1, s_2)^\mathsf{T}$, choose basic variables (x_1, x_2). Then

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

One way to obtain the inverse is to solve $Bq = v$:

$$2q_1 + q_2 = v_1, \quad q_1 + 2q_2 = v_2$$

$$\Rightarrow \quad q_1 = \frac{2v_1 - v_2}{3}, \quad q_2 = \frac{-v_1 + 2v_2}{3}.$$

Example: compute the point from the chosen columns

For $x = (x_1, x_2, s_1, s_2)^T$, choose basic variables (x_1, x_2). Then

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Set $s_1 = s_2 = 0$. The basic values are

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = B^{-1}\begin{pmatrix} 10 \\ 8 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \geq 0.$$

Thus the full BFS is $(4, 2, 0, 0)^T$, with objective 28. Neither plotted coordinate is zero; the zero coordinates here are the slacks.

Enumeration: six column selections in our example

All six pairs are independent: no two columns are proportional. Setting the other coordinates to zero gives

Basic variables	x_1	x_2	s_1	s_2	Feasible?	z
x_1, x_2	4	2	0	0	Yes	28
x_1, s_1	8	0	-6	0	No	–
x_1, s_2	5	0	0	3	Yes	25
x_2, s_1	0	4	6	0	Yes	16
x_2, s_2	0	10	0	-12	No	–
s_1, s_2	0	0	10	8	Yes	0

In general: first test column independence; then solve; then test nonnegativity. Compare objective values **only for feasible candidates**.

Why is looking at BFSs enough?

Fundamental existence theorem

For an equational LP with independent equality rows and nonnegative variables, if the feasible set is nonempty and the objective is **bounded above**, then an optimal BFS exists.

This gives a finite method: enumerate all bases, retain their BFSs, and compare objective values.

At most $\boxed{\binom{n}{m}}$ column selections.

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At most $\boxed{\binom{n}{m}}$ column selections.

Some selections are dependent or infeasible; several bases may give the same point.

The number grows quickly. Simplex uses the tableau to choose useful basis changes.

3. The simplex method

Improve a feasible basis, then certify the result

Simplex: begin with a feasible tableau

For our production model, the initial tableau is

$$\begin{array}{rcrrr} s_1 & = & 10 & -2x_1 & -x_2 \\ s_2 & = & 8 & -x_1 & -2x_2 \\ \hline z & = & 0 & +5x_1 & +4x_2 \end{array}$$

Set $x_1 = x_2 = 0$: basic values $(s_1, s_2) = (10, 8)$ are nonnegative.

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Choose x_1 to enter. With $x_2 = 0$,

$$s_1 \geq 0 \Rightarrow x_1 \leq 5, \qquad s_2 \geq 0 \Rightarrow x_1 \leq 8.$$

Thus s_1 **leaves** when $x_1 = 5$. The entering variable is a choice; the limiting rows determine the allowed step.

First pivot: rewrite every row

Isolate $x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$ and substitute:

$$\begin{array}{rcrrr} x_1 & = & 5 & -\frac{1}{2}x_2 & -\frac{1}{2}s_1 \\ s_2 & = & 3 & -\frac{3}{2}x_2 & +\frac{1}{2}s_1 \\ \hline z & = & 25 & +\frac{3}{2}x_2 & -\frac{5}{2}s_1 \end{array}$$

The current BFS is $(x_1, x_2, s_1, s_2) = (5, 0, 0, 3)$.

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Choose x_2 to enter and keep $s_1 = 0$. Then

$$x_1 \geq 0 \Rightarrow x_2 \leq 10, \qquad s_2 \geq 0 \Rightarrow x_2 \leq 2.$$

Thus s_2 **leaves** at $x_2 = 2$.

Second pivot: a tableau that proves optimality

Isolate $x_2 = 2 + \frac{1}{3}s_1 - \frac{2}{3}s_2$ and substitute:

$$\begin{array}{rcrrr} x_1 & = & 4 & -\frac{2}{3}s_1 & +\frac{1}{3}s_2 \\ x_2 & = & 2 & +\frac{1}{3}s_1 & -\frac{2}{3}s_2 \\ \hline z & = & 28 & -2s_1 & -s_2 \end{array}$$

The nonbasic variables are now s_1, s_2. Setting both to zero gives

$$\boxed{(x_1, x_2) = (4, 2), \qquad z^* = 28.}$$

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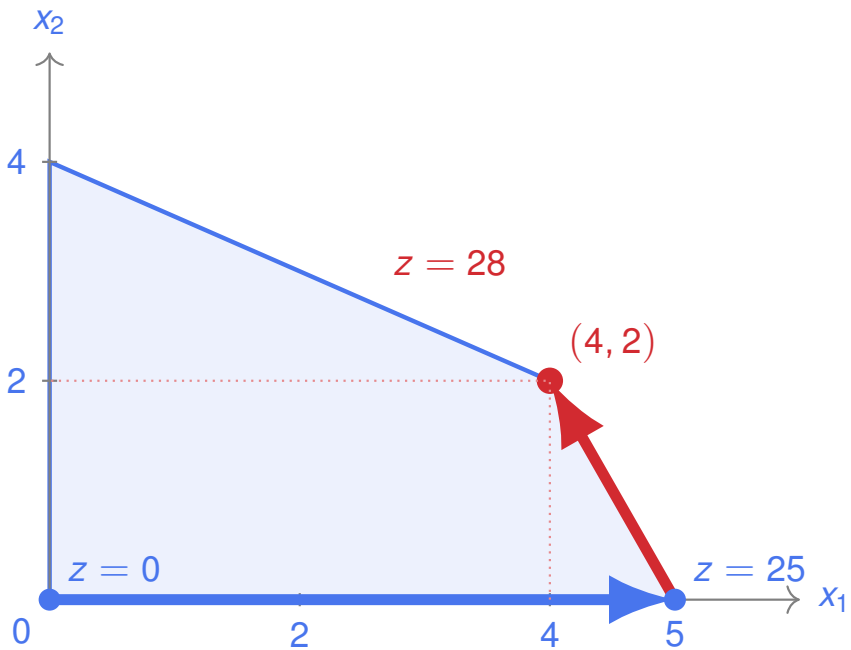
Every feasible plan has $s_1, s_2 \geq 0$. Consequently,

$$z = 28 - 2s_1 - s_2 \leq 28.$$

The final row proves a bound for **every feasible solution**, and the current BFS attains it.

Produce 4 units of P and 2 units of Q, for a daily profit of 2,800 MAD.

The same simplex moves in the plane



The two pivots move through

$$(0, 0) \longrightarrow (5, 0) \longrightarrow (4, 2).$$

Each move keeps feasibility and increases the objective.

The tableau records both the current BFS and its possible basic directions.

For a degenerate BFS, a pivot may change the basis without moving the point.

Read the general equation-style tableau

Split the columns into basic and nonbasic ones:

$$Bx_B + A_{\mathcal{N}} x_{\mathcal{N}} = b.$$

Then

$$\boxed{x_B = \underbrace{B^{-1}b}_{\bar{b}} - \underbrace{B^{-1}A_{\mathcal{N}}}_{\bar{A}} x_{\mathcal{N}},}$$

$$\boxed{z = \underbrace{c_B^{\mathsf{T}} B^{-1} b}_{z_0} + \sum_{j \in \mathcal{N}} r_j x_j, \qquad r_j = c_j - c_B^{\mathsf{T}} B^{-1} A_j.}$$

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Nonbasic terms are on the **right**: displayed basic-row coefficients are $-B^{-1}A_j$.

Every basic variable has reduced cost zero: $r_B^{\mathsf{T}} = c_B^{\mathsf{T}} - c_B^{\mathsf{T}} B^{-1} B = 0^{\mathsf{T}}$.

Basic directions connect the matrix and the pivot

Increase nonbasic variable x_j by $\theta \geq 0$ while the other nonbasic variables stay zero. The basic direction satisfies

$$d_j = 1, \quad d_i = 0 \ (i \in \mathcal{N} \setminus \{j\}), \qquad \boxed{d_B = -B^{-1}A_j}, \qquad Ad = 0.$$

The step must preserve $x_B + \theta d_B \geq 0$:

$$\theta_{\max} = \min_{k:\, d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}.$$

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At the first pivot's BFS, with basic variables (x_1, s_2), increasing x_2 gives

$$d_B = (-1/2, -3/2)^{\mathsf{T}}, \qquad \theta_{\max} = \min\{5/(1/2), 3/(3/2)\} = 2.$$

The gain per unit is $r_{x_2} = 3/2$: profit increases from 25 to $25 + 2(3/2) = 28$.

The two conditions for an optimal basis

For our maximization convention, a basis is optimal if

$$B^{-1}b \geq 0 \qquad \text{and} \qquad c^{\mathsf{T}} - c_B^{\mathsf{T}} B^{-1} A \leq 0^{\mathsf{T}}.$$

Optimality theorem

If the basic solution is feasible and every nonbasic reduced cost is nonpositive, that solution is optimal.

Conversely, if a BFS is optimal and **nondegenerate**, its nonbasic reduced costs are nonpositive.

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Conversely, if a BFS is optimal and **nondegenerate**, its nonbasic reduced costs are nonpositive.

Feasibility checks the **basic values**; optimality checks the **objective row**.

At a degenerate optimum, a particular basis may fail the reduced-cost sign test.

Two special cases of a simplex step

An unbounded improving direction

$$\begin{array}{rcl} x & = & 1 + s \\ \hline z & = & 1 + s \end{array} \qquad x, s \geq 0.$$

Increasing s raises profit and never makes x negative.

The objective is unbounded above.

A degenerate zero step For $x + s = 0$, maximize $z = x$:

$$\begin{array}{rcl} s & = & -x \\ \hline z & = & x \end{array} \qquad x, s \geq 0.$$

The basic value $s = 0$ blocks any positive increase in x.

The positive reduced cost does not give a positive feasible step.

An absence of a limiting row and a zero allowed step have different meanings.

4. Duality

From a feasible plan to a proof of its quality

Return to a feasible production plan

Let x_1, x_2 be the daily production amounts. Profit is measured in hundreds of MAD.

$$\begin{aligned} \max \quad & z = 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10, \\ & x_1 + 2x_2 \leq 8, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Consider $\widehat{x} = (3, 2)$.

$$\begin{aligned} 2(3) + 2 &= 8 \leq 10, \\ 3 + 2(2) &= 7 \leq 8. \end{aligned}$$

Its coordinates are nonnegative, so it is **feasible**.

$$z(\widehat{x}) = 5(3) + 4(2) = 23.$$

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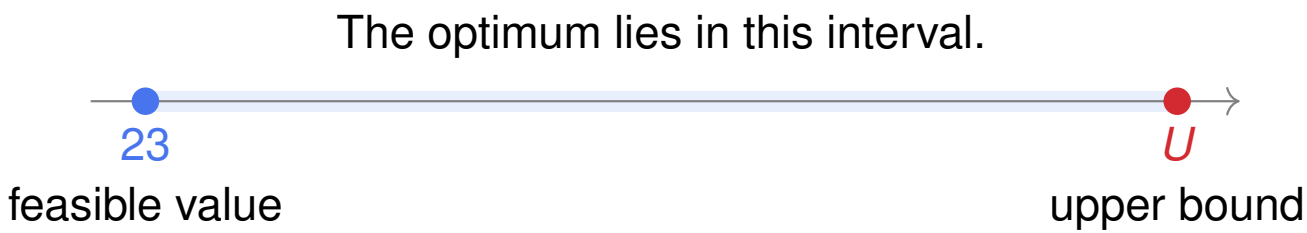
Feasibility proves that we can earn 23. How can we prove **how much more is possible**?

An upper bound limits the possible improvement

Let z^* be the best achievable profit. Our feasible plan gives $23 \leq z^*$.

Suppose we prove that **every feasible plan** earns at most U. Then

$$23 \leq z^* \leq U, \qquad 0 \leq z^* - 23 \leq U - 23.$$

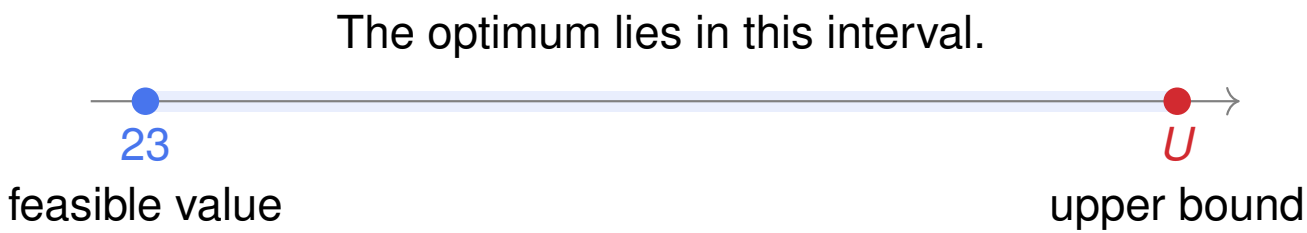


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If $U = 23$, the plan is optimal. If U is close to 23, it is close to optimal.

A positive gap bounds the possible loss; it does **not** by itself prove that the plan is suboptimal.

A first upper bound, by eye

Use the first resource inequality:

$$2x_1 + x_2 \leq 10 \quad \Longrightarrow \quad 4(2x_1 + x_2) = 8x_1 + 4x_2 \leq 40.$$

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Because $x_1 \ge 0$, every feasible plan satisfies

$$\underbrace{5x_1 + 4x_2}_{\text{profit}} \le 8x_1 + 4x_2 \le 40.$$

Thus

$$23 \le z^* \le 40, \qquad z^* - 23 \le 17.$$

The bound is valid even though we used only **one constraint**. We want a **smaller upper bound**.

A tighter bound: combine both resource limits

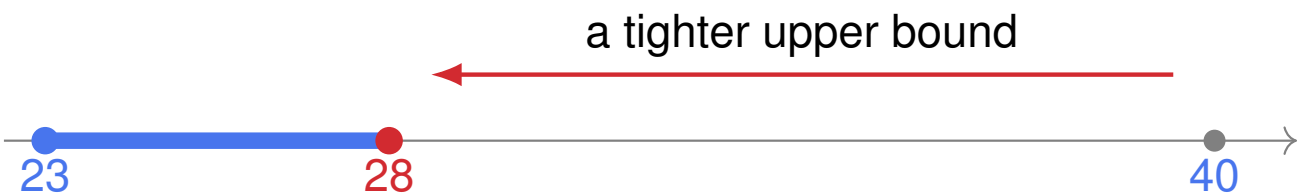
Multiply the first inequality by 2 and the second by 1, then add:

$$\begin{aligned} & 2(2x_1 + x_2) \leq 20, \\ + \quad & 1(x_1 + 2x_2) \leq 8 \\ \hline & 5x_1 + 4x_2 \leq 28. \end{aligned}$$

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$$0 \le z^* - 23 \le 5.$$

We improved the guarantee **without changing our feasible plan**.

Let the weights become decision variables

Different weights can produce different bounds:

Weight on resource 1	Weight on resource 2	Upper bound
4	0	40
1	3	34
2	1	28

For example, weights $(1, 3)$ give

$$5x_1 + 7x_2 \leq 34 \quad \Longrightarrow \quad 5x_1 + 4x_2 \leq 34.$$

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2	1	28

For example, weights $(1, 3)$ give

$$5x_1 + 7x_2 \leq 34 \quad \Longrightarrow \quad 5x_1 + 4x_2 \leq 34.$$

Call the unknown weights y_1, y_2.

Which weights make the bound **valid**, and which give the **smallest bound**?

Choose nonnegative weights, multiply, then add

Choose $y_1, y_2 \geq 0$ so multiplication preserves the inequality direction:

$$y_1(2x_1 + x_2) \leq 10y_1,$$
$$y_2(x_1 + 2x_2) \leq 8y_2.$$

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Add and collect the coefficient of each production variable:

$$\boxed{(2y_1 + y_2)x_1 + (y_1 + 2y_2)x_2 \leq 10y_1 + 8y_2.}$$

The right-hand side is our **candidate upper bound**.

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What conditions make the left-hand side at least the profit $5x_1 + 4x_2$?

Make the weighted sum dominate the objective

Since $x_1, x_2 \geq 0$, it is sufficient to impose

$$\underbrace{2y_1 + y_2}_{\text{coefficient of } x_1} \geq 5, \qquad \underbrace{y_1 + 2y_2}_{\text{coefficient of } x_2} \geq 4.$$

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For every feasible production plan, these conditions give

$$\begin{aligned} 5x_1 + 4x_2 &\leq (2y_1 + y_2)x_1 + (y_1 + 2y_2)x_2 \\ &\leq 10y_1 + 8y_2. \end{aligned}$$

The coefficients may be **larger** than 5 and 4; equality is not required.

We now have the constraints that make the weights valid certificates.

The search for the smallest bound is the dual LP

Primal: choose a production plan

$$\begin{aligned} \max \quad & z = 5x_1 + 4x_2 \\ \text{s.t.} \quad & 2x_1 + x_2 \leq 10, \\ & x_1 + 2x_2 \leq 8, \\ & x_1, x_2 \geq 0. \end{aligned}$$

A feasible plan gives a **lower bound** on the maximum profit.

Dual: choose constraint weights

$$\begin{aligned} \min \quad & w = 10y_1 + 8y_2 \\ \text{s.t.} \quad & 2y_1 + y_2 \geq 5, \\ & y_1 + 2y_2 \geq 4, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Feasible weights give an **upper bound** on the maximum profit.

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Feasible weights give an **upper bound** on the maximum profit.

The dual minimizes the bound subject to the conditions **we just derived**.

$$5x_1 + 4x_2 \leq z^* \leq 10y_1 + 8y_2.$$

The same construction in matrix notation

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$.

Primal

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax \leq b, \\ & x \geq 0. \end{aligned}$$

n variables; m constraint rows.

Dual

$$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & A^T y \geq c, \\ & y \geq 0. \end{aligned}$$

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m weights; n constraint rows.

One primal **row** gives one dual **variable**. One primal **variable** gives one dual **row**.

These counts exclude variable sign restrictions.

Why does the transpose appear?

The coefficient of x_j in the weighted sum of primal rows is

$$\boxed{(A^{\mathsf{T}} y)_j = \sum_{i=1}^{m} a_{ij} y_i .}$$

Read **column j of A** vertically, multiply by the row weights, and add.

Primal variable	Its column	Dual inequality
x_1	$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$	$2y_1 + y_2 \geq 5$
x_2	$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$	$y_1 + 2y_2 \geq 4$

Since $A^{\mathsf{T}} \in \mathbb{R}^{n \times m}$ and $y \in \mathbb{R}^m$, the vector $A^{\mathsf{T}} y$ has n entries, just like c.

Weak duality: every feasible pair gives a bound

Weak duality theorem

For every primal-feasible x and dual-feasible y,

$$\boxed{c^T x \leq b^T y.}$$

Weak duality: every feasible pair gives a bound

Weak duality theorem

For every primal-feasible x and dual-feasible y,

$$\boxed{c^T x \le b^T y.}$$

There are two inequality steps:

$$c^T x \le (A^T y)^T x = y^T A x \qquad \text{because } x \ge 0,\ A^T y \ge c,$$

$$y^T A x \le y^T b \qquad \text{because } y \ge 0,\ Ax \le b.$$

Thus a feasible pair traps the primal optimum:

$$c^T x \le z^* \le b^T y.$$

A matching pair certifies optimality

Primal candidate: $x = (4, 2)$

$$2(4) + 2 = 10 \leq 10,$$
$$4 + 2(2) = 8 \leq 8.$$

Both coordinates are nonnegative.

$$c^T x = 5(4) + 4(2) = 28.$$

Dual candidate: $y = (2, 1)$

$$2(2) + 1 = 5 \geq 5,$$
$$2 + 2(1) = 4 \geq 4.$$

Both weights are nonnegative.

$$b^T y = 10(2) + 8(1) = 28.$$

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Both weights are nonnegative.

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Weak duality gives $28 \leq z^* \leq 28$. Therefore **both candidates are optimal**.

Our original plan $(3, 2)$ is now proved to lose exactly $28 - 23 = 5$.

Strong duality: an exact certificate exists

Strong duality theorem

If the primal is **feasible** and its objective is **bounded above**, then both problems attain optimal solutions and

$$\max_{\substack{Ax \le b \\ x \ge 0}} c^T x = \min_{\substack{A^T y \ge c \\ y \ge 0}} b^T y.$$

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Weak duality: every feasible dual vector gives a valid upper bound.

Strong duality: among these bounds, an **exact one exists**.

A matching feasible pair proves optimality using weak duality alone. Strong duality guarantees that such a pair exists when the primal has a finite optimum.

Another source of bounds: relax the model

Removing constraints enlarges the feasible set. The **optimal value of the relaxed model** therefore bounds the original optimum.

Objective	Feasible plan gives	Relaxed optimum gives
Maximize	Lower bound	Upper bound
Minimize	Upper bound	Lower bound

In our example, dropping resource 2 gives an optimum of 40, attained by the relaxed plan $(0, 10)$.

$$z^* \leq 40.$$

That plan violates resource 2. It supplies a bound, not a feasible original decision.

Build a dual when the signs are mixed

These rules are for a **maximization primal** and a minimization dual.

Primal row $\longrightarrow$ **dual variable**	
$a_i^T x \leq b_i$	$y_i \geq 0$
$a_i^T x \geq b_i$	$y_i \leq 0$
$a_i^T x = b_i$	$y_i \in \mathbb{R}$

Choose the sign that turns a weighted row into an upper-bound inequality.

Primal variable $\longrightarrow$ **dual row**	
$x_j \geq 0$	$A_j^T y \geq c_j$
$x_j \leq 0$	$A_j^T y \leq c_j$
x_j unrestricted	$A_j^T y = c_j$

Require $c_j x_j \leq (A_j^T y) x_j$ for the allowed signs of x_j.

The objective is still $b^T y$. Alternatively, first transform the primal to canonical form.

Mixed signs: read each row and each column

Primal

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \\ & x_1 - x_2 = 1, \\ & x_1 \geq 0,\ x_2 \in \mathbb{R}. \end{aligned}$$

Dual

$$\begin{aligned} \min \quad & 4y_1 + y_2 \\ \text{s.t.} \quad & y_1 + y_2 \geq 3, \\ & y_1 - y_2 = 2, \\ & y_1 \geq 0,\ y_2 \in \mathbb{R}. \end{aligned}$$

Mixed signs: read each row and each column

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The second primal row is an equality, so y_2 **is unrestricted**.

The primal variable x_2 is unrestricted, so its dual row is an **equality**.

The coefficient of y_2 in the dual objective is the original right-hand side, 1.

Before complementary slackness: measure the margins

Return to the canonical pair. Assume **both x and y are feasible**.

Primal slack

$$s = b - Ax \geq 0.$$

For resource 1,

$$s_1 = 10 - (2x_1 + x_2).$$

It is unused capacity.

Dual slack

$$t = A^{\mathsf{T}} y - c \geq 0.$$

For the first dual row,

$$t_1 = 2y_1 + y_2 - 5.$$

It is the excess coefficient in our bound.

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It is the excess coefficient in our bound.

A constraint is **tight** when its slack is zero. A strict inequality has positive slack.

The gap is a sum of nonnegative terms

Insert $y^T Ax$ between the primal and dual objective values:

$$b^T y - c^T x = y^T (b - Ax) + x^T (A^T y - c)$$

$$= \sum_{i=1}^{m} \underbrace{y_i s_i}_{\geq 0} + \sum_{j=1}^{n} \underbrace{x_j t_j}_{\geq 0} .$$

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$$= \sum_{i=1}^{m} \underbrace{y_i s_i}_{\geq 0} + \sum_{j=1}^{n} \underbrace{x_j t_j}_{\geq 0}.$$

Every term is nonnegative because both candidates are feasible.

The gap is zero **exactly when every product is zero**.

$$\boxed{y_i s_i = 0 \text{ for all } i, \qquad x_j t_j = 0 \text{ for all } j.}$$

These are the **complementary slackness conditions**.

Complementary slackness: a test for an optimal pair

Complementary slackness theorem

For a primal-feasible x and a dual-feasible y, the two candidates are both optimal if and only if

$$\boxed{\begin{aligned} y_i\left[b_i - (Ax)_i\right] &= 0 \quad (i = 1, \ldots, m), \\ x_j\left[(A^{\mathsf{T}}y)_j - c_j\right] &= 0 \quad (j = 1, \ldots, n). \end{aligned}}$$

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At an optimal pair, strong duality makes the gap zero, so all products vanish.

Conversely, zero products make the objective values equal. Feasibility and weak duality then prove optimality.

Feasibility on both sides is essential.

Read the zero products one at a time

At an optimal primal–dual pair:

A primal row and its dual weight

$$y_i s_i = 0.$$

$$s_i > 0 \implies y_i = 0,$$

$$y_i > 0 \implies s_i = 0.$$

A positive weight requires a tight primal constraint.

An activity and its dual slack

$$x_j t_j = 0.$$

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$$x_j > 0 \implies t_j = 0,$$

$$t_j > 0 \implies x_j = 0.$$

A positive activity requires a tight dual constraint.

At least one factor must be zero; **both may be zero**. Tightness alone does not force the paired variable to be positive.

Find a certificate without guessing the weights

Take the primal-feasible candidate $x = (4, 2)$. Both production amounts are positive, so a matching dual certificate must satisfy

$$2y_1 + y_2 = 5, \qquad y_1 + 2y_2 = 4.$$

Find a certificate without guessing the weights

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Twice the first equation minus the second gives

$$3y_1 = 6 \quad \Longrightarrow \quad y_1 = 2, \qquad y_2 = 5 - 2(2) = 1.$$

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Verify before concluding. The vector $y = (2, 1)$ is nonnegative and satisfies both dual inequalities.

$$10(2) + 8(1) = 28 = 5(4) + 4(2).$$

Therefore both candidates are optimal.

Complementary slackness helps **find** candidates; feasibility and matching values **certify** them.

What can happen to a primal–dual pair?

For a maximization primal and its minimization dual:

If we know that...	We may conclude...
One has a finite optimum	Both have equal finite optimal values.
The primal is unbounded above	The dual is infeasible.
The dual is unbounded below	The primal is infeasible.
The primal is infeasible	The dual may be unbounded or infeasible.

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The primal is infeasible	The dual may be unbounded or infeasible.

Why the second row? A feasible dual vector would give a finite upper bound, contradicting an unbounded primal objective.

Infeasible does not automatically mean the other problem is unbounded.

5. Sensitivity analysis

Use an optimal tableau when the data change

Return to the production example

$$
\begin{aligned}
\max \quad & z = 5x_1 + 4x_2 \\
\text{s.t.} \quad & 2x_1 + x_2 \leq 10, \\
& x_1 + 2x_2 \leq 8, \\
& x_1, x_2 \geq 0.
\end{aligned}
$$

$$
\begin{array}{rcrrr}
x_1 & = & 4 & -\frac{2}{3}s_1 & +\frac{1}{3}s_2 \\
x_2 & = & 2 & +\frac{1}{3}s_1 & -\frac{2}{3}s_2 \\
\hline
z & = & 28 & -2s_1 & -s_2
\end{array}
$$

The optimal plan is $(x_1, x_2) = (4, 2)$, with $z^* = 28$. Basic variables are (x_1, x_2); nonbasic variables are (s_1, s_2). Profit values remain in **hundreds of MAD**.

We now change **one set of data at a time**. Can this basis still solve the modified problem?

Recover the inverse from the slack columns

The original slack columns form I_2. Hence the basic rows satisfy

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = B^{-1}b - B^{-1}\begin{pmatrix} s_1 \\ s_2 \end{pmatrix}.$$

The displayed coefficients of (s_1, s_2) are

$$-B^{-1} = \begin{pmatrix} -2/3 & 1/3 \\ 1/3 & -2/3 \end{pmatrix}.$$

Therefore

$$\boxed{B^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}}, \qquad c_B^{\mathsf{T}}B^{-1} = (5,4)B^{-1} = (2,1).$$

The sign change comes from putting the nonbasic variables on the right. Read the rows in the **specified basic-variable order**.

Local sensitivity: keep the two conditions valid

We start from an optimal basis. For the modified problem, we require

$$B^{-1}b \geq 0 \quad \text{and} \quad c^{\mathsf{T}} - c_B^{\mathsf{T}} B^{-1} A \leq 0^{\mathsf{T}}.$$

Changed data	What stays valid?	What needs checking?
New variable	Old plan with $u = 0$	New reduced cost
b changes	Reduced costs	New basic values
c changes	Basic values	New reduced costs

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$$B^{-1}b \geq 0 \quad \text{and} \quad c^{\mathsf{T}} - c_B^{\mathsf{T}} B^{-1} A \leq 0^{\mathsf{T}}.$$

Changed data	What stays valid?	What needs checking?
New variable	Old plan with $u = 0$	New reduced cost
b changes	Reduced costs	New basic values
c changes	Basic values	New reduced costs

These tests give the range where the **same basis stays optimal**. Outside that range, inspect and reuse the current tableau to reoptimize.

Case 1: add a new activity

Let $u \geq 0$ have resource column a and profit coefficient γ. The old plan remains feasible when $u = 0$.

$$x_B = B^{-1}b - B^{-1}A_{\mathcal{N}}x_{\mathcal{N}} - B^{-1}a\,u,$$

$$z = z_0 + \sum_{j \in \mathcal{N}} r_j x_j + \underbrace{\left(\gamma - c_B^{\mathsf{T}} B^{-1} a\right)}_{r_u} u.$$

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$$z = z_0 + \sum_{j \in \mathcal{N}} r_j x_j + \underbrace{\left(\gamma - c_B^{\mathsf{T}} B^{-1} a\right)}_{r_u} u.$$

If $r_u \leq 0$, the old basis remains optimal. If $r_u > 0$, inspect the allowed increase in u and pivot when possible.

The basis matrix remains $m \times m$: a new variable adds a column, not an equation.

New activity: one unit of each resource

Suppose $a = (1,1)^{\mathsf{T}}$. Then

$$B^{-1}a = (1/3, 1/3)^{\mathsf{T}}, \qquad r_u = \gamma - (2,1)a = \gamma - 3.$$

The updated tableau is

$$\begin{array}{rcrrrr} x_1 & = & 4 & -\frac{2}{3}s_1 & +\frac{1}{3}s_2 & -\frac{1}{3}u \\ x_2 & = & 2 & +\frac{1}{3}s_1 & -\frac{2}{3}s_2 & -\frac{1}{3}u \\ \hline z & = & 28 & -2s_1 & -s_2 & +(\gamma-3)u \end{array}$$

New activity: one unit of each resource

Suppose $a = (1, 1)^T$. Then

$$B^{-1}a = (1/3, 1/3)^T, \qquad r_u = \gamma - (2, 1)a = \gamma - 3.$$

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For $\gamma = 2$, the old plan stays optimal with $u = 0$.

For $\gamma = 4$, increasing u improves the objective. With $s_1 = s_2 = 0$, $x_2 = 2 - u/3$ reaches zero first, at $u = 6$.

New activity: pivot and check the result

Take $\gamma = 4$. Isolate u from the x_2 row:

$$u = 6 + s_1 - 2s_2 - 3x_2.$$

Substitute into the other rows:

$$\begin{array}{rcrrrr} x_1 & = & 2 & -s_1 & +s_2 & +x_2 \\ u & = & 6 & +s_1 & -2s_2 & -3x_2 \\ \hline z & = & 34 & -s_1 & -3s_2 & -3x_2 \end{array}$$

Set $s_1 = s_2 = x_2 = 0$. The new optimum is

$$\boxed{(x_1, x_2, u) = (2, 0, 6), \qquad z^* = 34.}$$

The basic values are nonnegative and all nonbasic reduced costs are negative.

Case 2: change a resource amount

Let the first resource change from 10 to $10+\delta$:

$$b' = \begin{pmatrix} 10+\delta \\ 8 \end{pmatrix}, \qquad B^{-1}b' = \begin{pmatrix} 4+\frac{2}{3}\delta \\ 2-\frac{1}{3}\delta \end{pmatrix}.$$

The reduced costs are still $-2, -1$, because A and c are unchanged. Thus the old basis remains optimal exactly when

$$4+\tfrac{2}{3}\delta \geq 0, \quad 2-\tfrac{1}{3}\delta \geq 0 \iff \boxed{-6 \leq \delta \leq 6.}$$

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The reduced costs are still $-2, -1$, because A and c are unchanged. Thus the old basis remains optimal exactly when

$$4+\tfrac{2}{3}\delta \geq 0, \quad 2-\tfrac{1}{3}\delta \geq 0 \quad \Longleftrightarrow \quad \boxed{-6 \leq \delta \leq 6.}$$

Within that range,

$$z^*(\delta) = 5(4+\tfrac{2}{3}\delta) + 4(2-\tfrac{1}{3}\delta) = 28+2\delta.$$

For $\delta = 3$, the optimal production changes to $(6,1)$ and profit to 34, even though the **basis stays the same**.

Beyond the resource range: reuse the tableau

For $\delta = 9$, the first capacity is 19. The old basic values become $(10, -1)$, so this basis is infeasible. Its second row is

$$x_2 = -1 + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2 \quad \Rightarrow \quad s_1 = 3 + 3x_2 + 2s_2.$$

Substitute to obtain

$$\begin{array}{rcrrr} x_1 & = & 8 & -2x_2 & -s_2 \\ s_1 & = & 3 & +3x_2 & +2s_2 \\ \hline z & = & 40 & -6x_2 & -5s_2 \end{array}$$

Setting $x_2 = s_2 = 0$ gives a feasible optimum: $(x_1, x_2) = (8, 0)$, $z^* = 40$.

The local formula $28 + 2\delta$ applies only in its valid range.

Case 3: change a unit profit

Let the profit coefficient of x_1 change from 5 to κ. The feasible set and the basic values $(4, 2)$ stay unchanged.

Substitute the basic rows into the **new objective**:

$$\begin{aligned} z &= \kappa\left(4 - \tfrac{2}{3}s_1 + \tfrac{1}{3}s_2\right) + 4\left(2 + \tfrac{1}{3}s_1 - \tfrac{2}{3}s_2\right) \\ &= 4\kappa + 8 + \frac{4 - 2\kappa}{3}s_1 + \frac{\kappa - 8}{3}s_2. \end{aligned}$$

The old basis stays optimal exactly when

$$\boxed{2 \le \kappa \le 8.}$$

For $\kappa = 6$, the optimal plan is still $(4, 2)$, but profit becomes 32. At the endpoints, another plan can also be optimal.

Beyond the profit range: use the positive reduced cost

For $\kappa = 9$, the updated objective row is

$$z = 44 - \tfrac{14}{3} s_1 + \tfrac{1}{3} s_2.$$

The old plan is feasible but not optimal. Let s_2 enter. With $s_1 = 0$, $x_2 = 2 - \tfrac{2}{3} s_2$ reaches zero at $s_2 = 3$.

Beyond the profit range: use the positive reduced cost

For $\kappa = 9$, the updated objective row is

$$z = 44 - \tfrac{14}{3} s_1 + \tfrac{1}{3} s_2.$$

The old plan is feasible but not optimal. Let s_2 enter. With $s_1 = 0$, $x_2 = 2 - \tfrac{2}{3} s_2$ reaches zero at $s_2 = 3$.

After pivoting,

$$\begin{array}{rcrrr} x_1 & = & 5 & -\tfrac{1}{2} x_2 & -\tfrac{1}{2} s_1 \\ s_2 & = & 3 & -\tfrac{3}{2} x_2 & +\tfrac{1}{2} s_1 \\ \hline z & = & 45 & -\tfrac{1}{2} x_2 & -\tfrac{9}{2} s_1 \end{array}$$

Thus the new optimal production is $(5, 0)$, with profit 45. The old plan's profit under this new objective was 44.

What does "unchanged" actually mean?

Modification	Same optimal basis?	Same production?	Same profit?
New activity, $\gamma = 2$	Yes	Yes, with $u = 0$	Yes
Capacity, $\delta = 3$	Yes	No: $(6, 1)$	No: 34
Profit, $\kappa = 6$	Yes	Yes: $(4, 2)$	No: 32

An optimal basis identifies **which variables are basic**. Their values depend on b; their objective coefficients depend on c.

Always state separately the modified optimal plan and its modified objective value.

Match the question to the method

What is requested?	What your answer should contain
Formulate a model	Variables, units, objective, constraints, domains
Solve graphically	Feasible region, labeled vertices, objective, conclusion
Enumerate BFSs	Independent selections, basic values, feasibility, values
Use simplex	Tableaux, entering/leaving choices, final certificate
Use duality	Dual formulation, feasibility, matching bound
Analyze a change	Updated basis tests, valid range, new plan and value

A complete answer states **why** the displayed equations justify the conclusion.

The main ideas to keep together

1. **Model first.** Decide what varies, what is allowed and what is optimized.
2. **Represent carefully.** Variable order, units and inequality directions matter.
3. **Solve with a justification.** A feasible plan needs an optimality certificate.
4. **Read the tableau.** Basic values describe feasibility; reduced costs describe gains.
5. **Update what changed.** Reuse the optimal basis while its two conditions hold.

Write the conclusion in terms of the **original decision variables and units**.